MATHEMATICS
Modern Algebra I
Excercises 3. Due in class September 17.

(1) Let G be a group, N, H sugroups such that N is normal in G. Show that a) conjugation induces a homomorphism $f : H \to Aut(N)$; b) if $N \cap H = \{1\}$, then the map $N \times H \to NH$ is a bijection which is an isomorphism if and only if f is trivial.
(2) Let N, H be groups and $f : H \to Aut(N)$ be a given homomorphism. Show that $G = N \times H$ with multiplication given by $(n, h)(n', h') = (n \cdot f(h)(n'), hh')$ is a group (the semidirect product of N, H) and N is a normal subgroup of G.
(3) Classify all groups of order 55.
(4) Show that for any prime number p, all groups of order p^2 are abelian.
(5) If H, K are solvable subgroups of a group G with H normal in G, show that HK is a solvable subgroup of G.
(6) Does there exist a surjective homomorphism $S_4 \to S_3$?
(7) Find a p-Sylow of S_6 for $p = 2, 3$.
(8) Show that there are exactly two non-isomorphic groups of order 8.
(9) Show that if G is a group of order 40 (respectively 12), then G has a normal Sylow subgroup.